

ORIGINAL ARTICLE

Targeted (PCR-based) screening of antibiotic resistance genes' prevalence in the gut microbiota of tribal people of Nabarangpur, Odisha, India

M. Sethi^{1,2}, V. Mohanty¹, S. Mishra^{1,2}, J. Dash¹, S. Mahapatra¹, D. Parida^{1,2}, A.P. Mohapatra^{1,2}, R. Mishra¹, P. Prasad¹, A. Parida¹ and S. Senapati¹ 

¹ Institute of Life Sciences, Bhubaneswar, Odisha, India

² Regional Centre for Biotechnology, Faridabad, Haryana, India

Significance and Impact of the Study: This is the first attempt to check the prevalence of different ARGs in the gut microbiota of tribal people of Nabarangpur, Odisha, which is a small contributing step in controlling the evolving antibiotic resistance in India. The results of this study showed the dominance of tetracycline resistance genes (*tetW*, *tetQ* and *tetM*) as well as the similar pattern of distribution of ARGs among the three major tribes of this location. These findings provide a reference to monitor the evolution of antimicrobial resistance in this population in the future.

Keywords

antibiotic resistance, ARGs, faecal DNA, PCR, public health, tribal.

Correspondence

Ajay Parida and Shantibhusan Senapati, Institute of Life Sciences, Nalco Square, Bhubaneswar-751023, Odisha, India.

E-mail: drjayparida@gmail.com;

senapati@ils.res.in

Manisha Sethi and Varshasruti Mohanty contributed equally.

2022/0248: received 22 September 2021, revised 21 December 2021 and accepted 23 December 2021

doi:10.1111/lam.13642

Abstract

Antibiotic resistance is a major public health concern worldwide. The gut microbiota harbours multiple antibiotic resistant genes (ARGs) that contribute to the existing and future microbial population in a community or ecosystem. This study aimed to investigate the prevalence of 35 antibiotic resistance genes (ARGs) in the gut microbiota of the tribal people of Nabarangpur, Odisha, India. A total of 83 faecal samples were collected from three different tribes (Bhatra, Gond, and Paraja). Total faecal DNA was extracted, and the simplex polymerase chain reaction was performed to detect selected ARGs. Further analysis was done to estimate the incidence of these ARGs across these tribes based on alcohol consumption habits. We identified a higher prevalence of tetracycline resistance genes (*tetW*, *tetQ* and *tetM*) in the gut microbiota among three populations. Furthermore, a significant ($P = 0.024$) difference in ARG prevalence against vancomycin in individuals with and without alcohol consumption habits was noticed. The overall distribution of ARGs among the three major tribes of this location was found to be very similar. Together, irrespective of the tribes, the people of this location have gut microbiota harbouring different kinds of ARGs and tetracycline-resistant genes are the most commonly found ARGs.

Introduction

The human body is an example of an open ecosystem, which harbours beneficial and harmful microbes. The gut is the largest reservoir of microbes and has received the most attention fuelled by increasingly accessible DNA-sequencing platforms that elucidate the fundamental role of the gut microbiota in the host, and possible avenues for modifications. The interesting findings obtained using

these platforms suggest that the gut microbiota can easily be affected by various factors such as aging, diet, mental stress, poor lifestyle, use of antibiotics, infections, environmental exposure and topographical location (Ghosh *et al.* 2014). It was observed that the commensal and enteric pathogens living in the gut of healthy individuals harbour several antimicrobial resistance genes (ARGs). The ARGs are also linked to mobile genetic elements, and the resistance function can be disseminated to the sensitive

microbial population through horizontal gene transfer (Bag *et al.* 2019).

Antibiotics were introduced to the world as a blessing through the discovery of penicillin by Alexander Fleming in 1928, which was a grace to the medicinal era (Sengupta *et al.* 2013). Later, antibiotics were used in the food industry and both in the animal and agricultural industries as growth-promoting agents (Van *et al.* 2020). Frequent use of antibiotics in clinics, agriculture, and the food industry has made some bacteria resistant to multiple antibiotics (Moriguchi *et al.* 2020). Antibiotic resistance, a bacterial defence mechanism that protects it against specific antibiotics, may be present intrinsically or can be acquired from the external environment. Acquisition of intrinsic antibiotic resistance develops naturally by the microbes producing antibiotics for its defence and sustenance in a competitive environment, whereas, acquired antibiotic resistance develops mainly in pathogens by either chromosomal mutation or a horizontal gene transfer from other bacteria. Any mutation in intrinsic and acquired antibiotic resistance can contribute to the emergence of new drug-resistant pathogens (Cockerill 1999; Rolain 2013). So far, conventional/culture-based methods (such as disc diffusion, microdilution, agar dilution method) and molecular-based methods (such as PCR-based method, metagenomics studies/techniques) have been developed for the detection of antibiotic resistant genes (Cockerill 1999). PCR-based detection of antibiotic resistance genes is one of the simplest methods and easy to analyse (Ohashi and Fujisawa 2017). It also has the potential to be adopted as a screening strategy for a larger population.

Antibiotic resistance is rapidly evolving in India. There is a regional variation in the incidence rate. Based on the report by World Health Organization in 2019, the reduction of excessive and inappropriate use of antimicrobials in patient treatment, agriculture, and animal farming is a public health priority. Nevertheless, treatment is necessary in many cases, raising the question of how to treat effectively with antibiotics while simultaneously reducing the risk of antibiotic resistance (Angst *et al.* 2021). It is, therefore, always necessary to screen the resistant bacteria in the human host, to elucidate their transfer mechanism, and their further spread into different hosts through environmental factors. There is not much attempt to understand these ARG distributions in the most primitive tribal populations in India owing to an assumption that their limited exposure to antimicrobial drugs might have protected them to acquire microbes with ARGs. However, in the view of public health concern, periodic monitoring of the prevalence of ARGs in the tribal population will help to improve their livelihood and health by providing supporting evidence to the government and policy-makers.

Here, we have investigated the presence of ARGs in the stool samples collected from three remote tribes—Bhatra, Gond, and Paraja of Nabarangpur district of Odisha, India, with no previous report of exposure to any antibiotics for the last 3 months. Nabarangpur district is situated in the greenery of south-western Odisha. Most of its population is tribal, and the land is heavily forested and has some low hills. Different tribes residing in Nabarangpur have their own culture, language, and lifestyle. In this area, the government has provided health care centers at different locations. However, the distance between these centers/hospitals varies for different villages. The majority of this population primarily consult local quacks for common symptoms like diarrhoea, vomiting, and fever and take local medicines (mostly herbal products). The common antibiotics that are mostly consumed by this population include amoxicillin, cefadroxil, metronidazole, and ofloxacin. Nabarangpur, an overwhelming agrarian region, has over 90% of its occupants relying upon cultivation for their livelihood. Here, we have investigated the presence (/absence) of important ARGs in the total DNA isolated from the stool samples of tribal people of Nabarangpur district, Odisha, by simplex PCR-based method using ARG-specific primers.

Results and discussion

A total of 83 volunteers belonging to three different tribes—Paraja, Gond, and Bhatra—from three different locations of Nabarangpur district (Maidalpur, Malekot, and Jharigaon), Odisha, India, were included in this study (Fig. 1, Table S1). Simplex PCR-based analysis identified the presence of multiple ARGs for antibiotics such as macrolide, tetracycline, carbapenem, fluoroquinolone, ampicillin, streptomycin, sulphonamide, and vancomycin.

However, in this study, we did not notice any positive samples for certain antibiotic-specific ARGs like genes known for Methicillin resistance (Table 1, Table S2). On average, seven antibiotic resistance genes were detected per individual (Table S2).

Although the importance of the gut microbiota to human health has been studied for decades, few studies have looked at the impact of sex on the gut microbiota in human intestines (Ding and Schloss 2014; Haro *et al.* 2016; Sinha *et al.* 2019; Takagi *et al.* 2019). However, the findings of each investigation on the differences in microbial taxa between men and women are contradictory. The outcomes of Asian studies were not the same as those of Western ones (Kim *et al.* 2020). In the current study, no correlation of age, or sex with ARGs was observed (data not shown). In particular, *tetW* and *tetQ* were found in faecal samples of 97.59% (81/83) and 95.18% (79/83) individuals, respectively. In contrast, the prevalence of

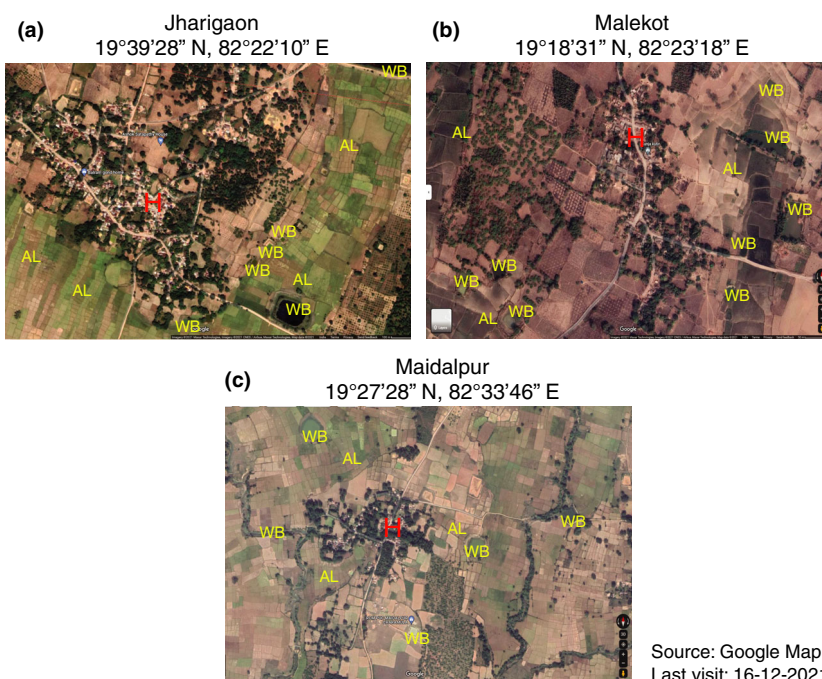


Figure 1 Topographical view of sampling sites at Nabarangpur, Odisha, India (H—household, AL—agricultural land, WB—water body). Source: Google Map, Last visit: 16.12.2021.

msr A/B (1/83), *mphA* (6/83), *ereA* (3/83), *ermA* (3/83) and *tetE* (2/83) were lower in human faecal samples. In addition, antibiotic resistance genes against tetracyclines-*tetM*, *tetQ* and *tetW*; carbapenem *bla_{OXA-48}* and sulfonamide-*sulII* were most commonly detected in faecal bacterial genomes. The other antibiotic resistance genes against vancomycin (*vanA* and *vanC*), macrolides (*ereA*, *ermA*, *ermC*, *mphA* and *msrA/B*), fluoroquinolones (*qnrA*, *qnrD* and *qnrS*), ampicillin (*ampC*), streptomycin (*aadA* and *aadE*) and tetracyclines (*tetC*, *tetE*, *tetK* and *tetT*) identified in this study, were detected in less than 20% of subjects. These results indicate that antibiotic resistance genes against carbapenem, sulfonamide and tetracyclines are likely to be widespread in the faecal/gut-associated bacteria of the tribal population of Nabarangpur.

Both host and environmental factors play major roles in shaping the gut microbiota of individuals (Hasan and Yang 2019). Moreover, due to the strict tradition of marriage within the same caste/tribe, we assume that the genetic makeup of these three tribes might have certain differences. Based on these backgrounds, we were interested in evaluating the distribution pattern and relative abundance of different ARGs in the studied populations. Overall, the pie chart results showed the presence of different ARGs at different percentages across these three tribes, and ARGs against tetracyclines were found most dominant among them (Fig. 2). However, the

distribution patterns of ARGs are similar among different tribes (Fig. 2). This could be since they share a similar ecosystem. Hence most of the microbiota colonized in their gut might have similar compositions. We identified a significant presence of ARGs against carbapenem in the Bhatra tribe ($P = 0.039$) and Paraja tribe ($P = 0.013$) compared with the Gond tribe (Fig. 3a). On the contrary, we observed a higher presence of ARGs against fluoroquinolones in the Gond tribe ($P = 0.05$) compared with the Paraja tribe (Fig. 3b). We also identified a significant presence of ARGs against streptomycin (Fig. 3c) and macrolides (Fig. 3d) among the Bhatra tribe compared with the Paraja tribe.

In the target population, we noticed 51 and 32 numbers of individuals with and without alcohol consumption habits, respectively (Fig. 4a, Table S3). Chronic alcohol consumption is known to have an impact on the composition of gut microbiota (Ciocan *et al.* 2018). Hence, we hypothesized that in our target population, alcohol-consuming and non-consuming individuals might have different patterns of gut microbes and thus have a differential distribution of ARGs. Furthermore, the results showed a significant difference ($P = 0.047$) between the presence of 35 ARGs with the alcohol consumption status (Fig. 4b). The statistical analysis of the PCR results revealed that the intake of alcohol significantly decreased the abundance of ARGs (Fig. 4b). Interestingly, we

Table 1 Results of PCR-based screening of 35 ARGs carried out in faecal sample collected from the tribal people of Nabarangpur, Odisha (*n* = 83)

Antibiotic	Resistant gene	Number of positive subjects	% of positive subjects
Macrolide	<i>ermA</i>	3	3.61
	<i>ermB</i>	22	26.51
	<i>ermC</i>	4	4.82
	<i>ermT</i>	14	16.87
	<i>ereA</i>	3	3.61
	<i>msrA/B</i>	1	1.20
	<i>mphA</i>	6	7.23
Tetracycline	<i>tetA</i>	46	55.42
	<i>tetB</i>	20	24.10
	<i>tetC</i>	7	8.43
	<i>tetD</i>	12	14.46
	<i>tetE</i>	2	2.41
	<i>tetG</i>	5	6.02
	<i>tetK</i>	6	7.23
	<i>tetL</i>	11	13.25
	<i>tetM</i>	78	93.98
	<i>tetP</i>	23	27.71
	<i>tetQ</i>	79	95.18
	<i>tetS</i>	30	36.14
	<i>tetT</i>	5	6.02
	<i>tetW</i>	81	97.59
Fluoroquinolone	<i>qnrA</i>	5	6.02
	<i>qnrS</i>	15	18.07
	<i>qnrD</i>	6	7.23
	<i>aac(6')-Ib-cr</i>	20	24.10
Carbapenem	<i>bla_{OXA-48}</i>	55	66.27
Ampicillin	<i>ampC</i>	5	6.02
Methicillin	<i>mecA</i>	0	—
Streptomycin	<i>aadA</i>	11	13.25
	<i>aadE</i>	2	2.41
Sulfonamide	<i>sulI</i>	28	33.73
	<i>sulII</i>	39	46.99
Vancomycin	<i>vanA</i>	10	12.05
	<i>vanB</i>	20	24.10
	<i>vanC</i>	9	10.84

identified a significant presence of ARGs against vancomycin in the gut microbiome of nonalcoholic people compared with the alcoholic group (Fig. 4c). The abundance of other ARGs except vancomycin did not show any significant difference among the alcoholic and non-alcoholic populations. Further studies in a larger cohort are warranted to validate these findings.

Antibiotic resistance has been observed in both humans and animals belonging to rural/remote areas with limited or no exposure to antibiotics (Souza *et al.* 1999; Bartoloni *et al.* 2004). In 2004, a study by Rossolini *et al.*, demonstrated the incidence of acquired antibiotic resistance in commensal *Escherichia coli* isolates in a remote Guarani

Indians Bolivian community with minimal exposure to antibiotics (Bartoloni *et al.* 2004). The community has limited access to outsiders and antibiotics where the rain-water was the only source of water (Bartoloni *et al.* 2004). The findings of our study also showed that the tribes of the Nabarangpur district of Odisha, India, also harbour gut microbes with different ARGs in their genome.

Various metagenomic studies from sewage, water, animals, and human faeces, etc. have confirmed the presence of a large reservoir of new ARG subsets collectively known as intrinsic resistome (Hu *et al.* 2013; Rolain 2013; Karkman *et al.* 2019). Hu *et al.* (2013) had identified 1093 ARGs in the human gut microbiome by a metagenomic study of a large cohort from three different countries, China, Denmark, and Spain, and confirmed a high abundance of tetracycline resistance genes. Chng *et al.* (2020) reported the importance of characterizing intrinsic resistome in hospitals to target resources for preventing infections. In a recent study, Milanovic *et al.* showed the abundance of ARGs against tetracycline (*tetW* and *tetM*) and erythromycin (*ermB*) in the Italian population. According to the Global Action Plan on Antimicrobial Resistance, screening of ARGs is one of the key objectives for minimizing the incidence of antibiotic resistance (Milanović *et al.* 2020). Previous studies have demonstrated the presence of antibiotic resistance genes against macrolides, tetracyclines, β -lactams, fluoroquinolones, vancomycin and sulphonamide using metagenomics, and microarray analysis (Seville *et al.* 2009; Sommer *et al.* 2009; Hu *et al.* 2014; van Schaik. 2015). Thus, there is a requirement for monitoring ARGs at the environment and community level for understanding the multidrug-resistance evolution.

In this study, we could find an interesting correlation that alcohol consumption leads to less occurrence of ARGs for vancomycin. As there is no supportive study corroborating the idea that alcohol consumption may directly induce acquisition of any ARGs, so it can be inferred that the differential gut microbiota of these populations may be a responsible factor. It is known that some commensal bacteria in the human gut like genus *Lactobacillus* shows intrinsic resistance to vancomycin (Gueimonde *et al.* 2013) and the assumption cannot be denied that those individuals possessing higher ARGs for vancomycin might harbour more of those groups of bacteria dominating in their gut flora.

Although this study provides the first comprehensive ARG screening in this population's gut microbiota, we also acknowledge certain limitations in the current study. First, the limited sample size from the three tribes might not be the true representative of the entire tribal population despite having the same geographical

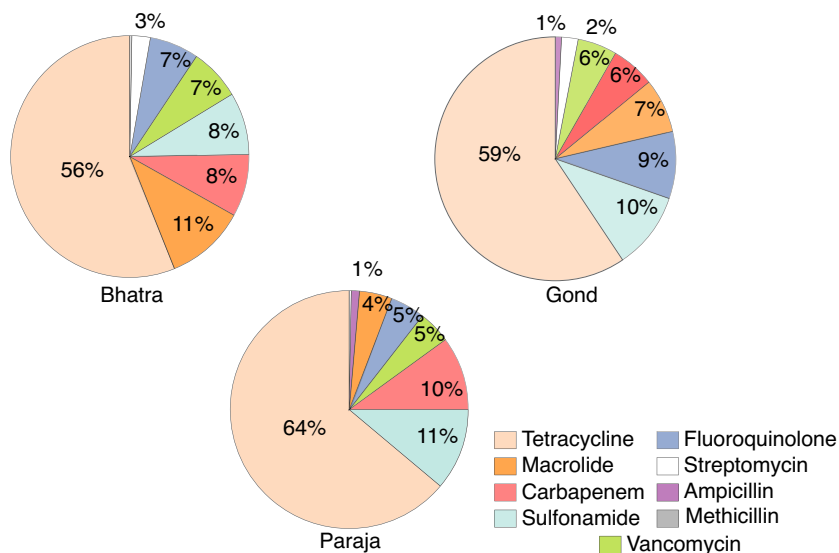


Figure 2 Distribution of antibiotic resistance genes across the three ethnic tribes, Bhatra, Gond and Paraja of Nabarangpur, Odisha.

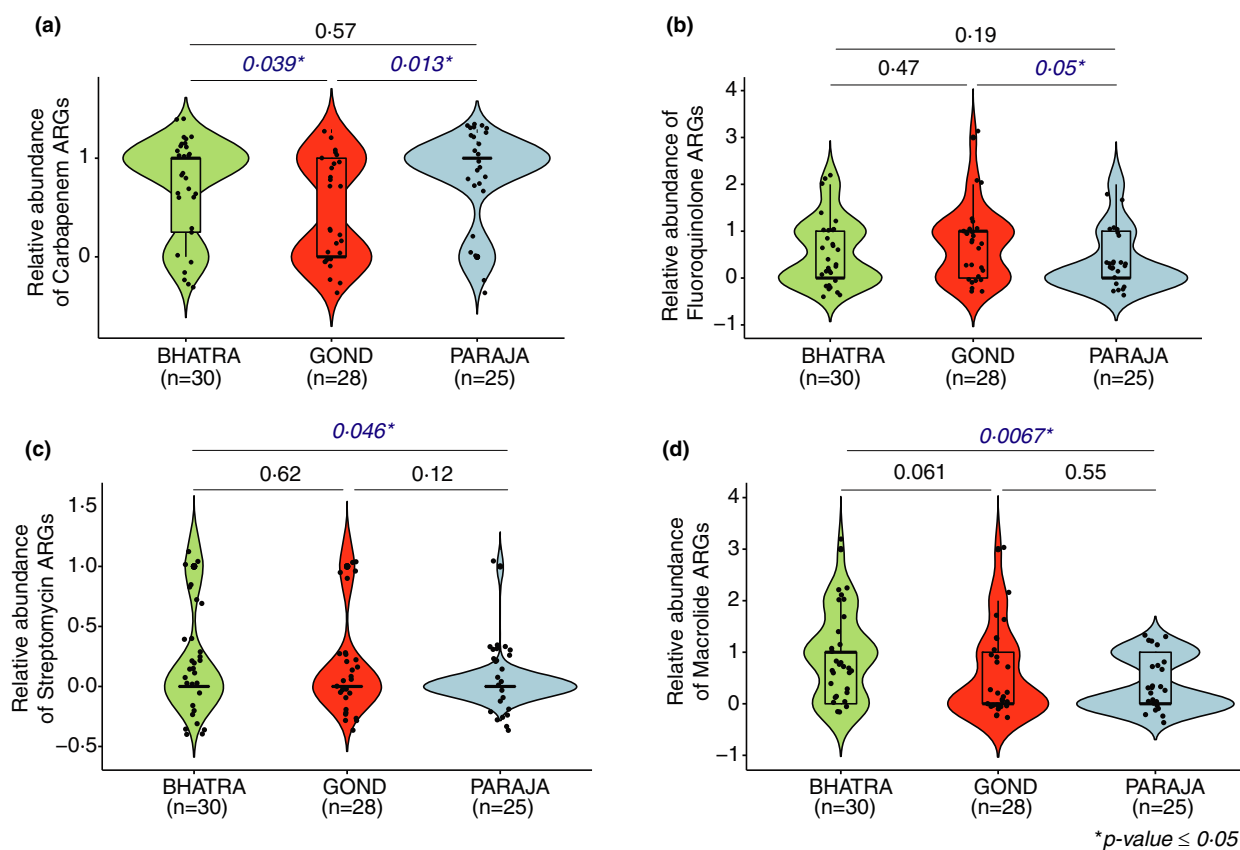


Figure 3 Relative abundance of antibiotic resistance genes against (a) carbapenem, (b) fluoroquinolone, (c) streptomycin, and (d) macrolide in the Bhatra, Gond, and Paraja tribes of Nabarangpur, Odisha.

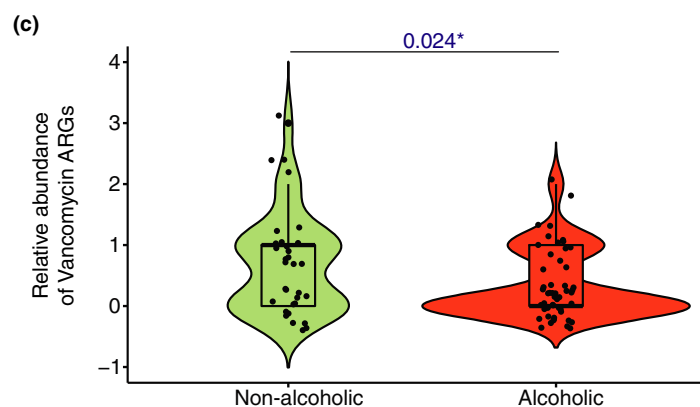
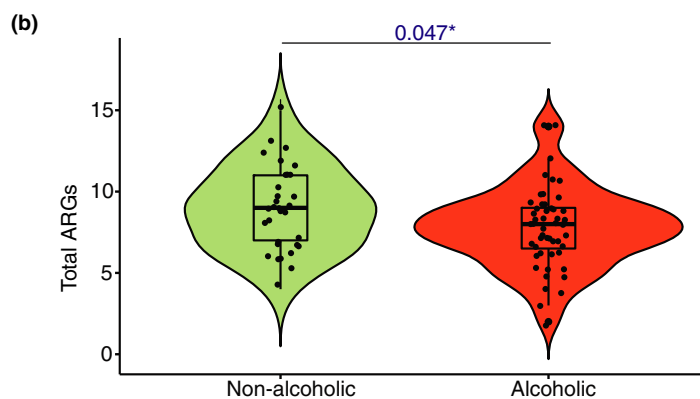
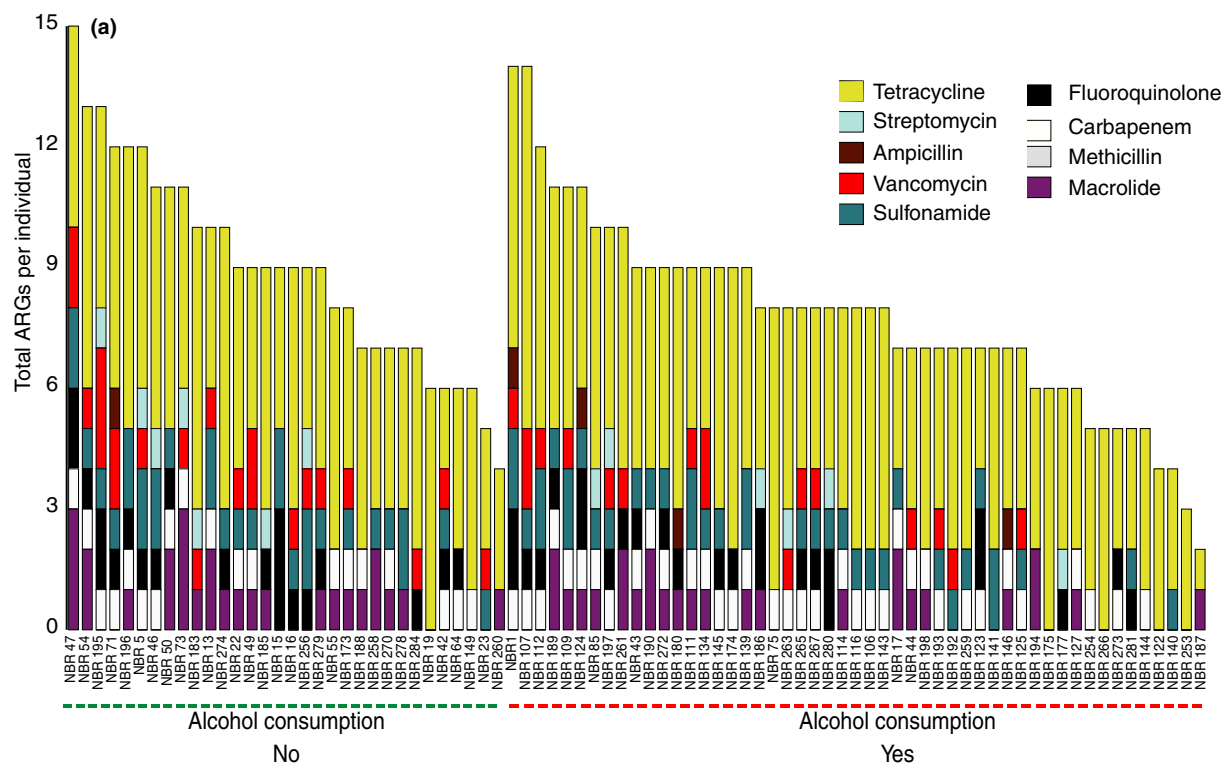


Figure 4 (a) Distribution of antibiotic resistance genes across the three tribes namely Bhatra, Gond, and Paraja of Nabarangpur, Odisha, segregated based on alcohol consumption. (b) Relative abundance of 35 antibiotic resistance genes in an alcoholic and nonalcoholic population of Bhatra, Gond and Paraja tribes of Nabarangpur, Odisha; (c) Significant ($P \leq 0.05$) prevalence of antibiotic resistance gene against vancomycin in alcoholic and nonalcoholic population of the Bhatra, Gond and Paraja tribes of Nabarangpur, Odisha

location. Second, the uneven distribution of age, sex, and tribe numbers has limited our scope for correlation studies in a more detailed manner. The targeted ARG analysis through the PCR method also has its inherent limitations like biases towards only specific genes and lack of global ARG information. Again, it will be worthwhile to explore and screen the same ARGs in different disease states.

Together, it can be concluded that the use of antibiotics is not the only reason for the presence or absence of ARGs in these tribal people as these people had no medical history of any antibiotic consumption, at least 3 months before the collection of samples. No major difference in the ARG distribution among tribes indicates that the host factors might have less contribution than the environment in establishing the gut microbes of this population. During the ongoing COVID-19 pandemic, antibiotics have been rampantly used to control secondary infections in SARS-CoV-2-infected patients. Unfortunately, in many parts of the world including India due to ignorance, antibiotics have been wrongly used by a large number of the population without any formal medical guidance. The rational and irrational use of antibiotics is believed to be a major concern in the emergence of new antibiotic-resistant bacteria in the post-COVID-19 scenario. We believe that our findings will provide a reference to compare the post-COVID-19 with pre- or early COVID-19 ARGs in this population. Moreover, the information obtained from this study will have further implications in designing targeted interventions to reduce exposure, persistence, and transmission of antibiotic resistance.

Materials and methods

Ethics approval and consent

The current study was done as a part of the ongoing tribal health and nutrition flagship program of the Institute of Life Sciences, Bhubaneswar. This study was approved by the Institutional Human Ethical Committee (IHEC), Institute of Life Sciences, Bhubaneswar, India (IEC Ref No: 87/HEC/19). Samples were collected from all the volunteers with their consent to publish the results of this study while keeping their identities confidential. They were informed about the importance of this study and the appropriate institutional forms have been archived.

Recruitment of volunteers

In the current study, a total of 83 volunteers belonging to three different tribes Paraja, Gond, and Bhatra were randomly selected irrespective of their age, sex, and food habits from three different locations of Nabarangpur district (Maidalpur, Malekot, and Jharigaon), Odisha. All the volunteers belonging to these three tribes have limited exposure to antibiotics and entirely rely on agriculture for their livelihood. Out of 83 volunteers, 25 were from the Paraja tribe, 28 were from the Gond tribe and 30 were from the Bhatra tribe. The volunteers included in this study were healthy with self-reported no medical history of chronic illness and/or major surgery. The only exclusion criteria was antibiotic intake in the last 3 months before sample collection.

Sample collection

Prior to sample collection, all the participants were oriented about the procedure of collecting samples without soil and/or environmental contamination of the faecal samples. The samples from 83 healthy volunteers were collected from three different tribes in a sterile DNA/RNA Shield Faecal collection tube (Zymo Research—R1101) individually on December 2020. The volunteers were instructed to collect about one scoop of faecal matter in the collection tube and mixed immediately, and the tubes were kept in cool packs within 30 min of defecation. The faecal samples were transported to ILS laboratory in a refrigerator. After reaching the Laboratory, the samples were stored at 4°C till further processing. Information related to their height, weight, medical history, food, and personal details were collected and provided in Table S1. The volunteers were further categorized into two groups based on the alcohol consumption habit.

Isolation of faecal bacterial DNA

The bacterial DNA was extracted manually from each stool sample according to the Translational Health Science and Technology Institute (THSTI) method of High-Quality Metagenomics DNA Extraction from Human and Environmental Samples and eluted in 60 µl of TE (Tris-EDTA) buffer (Bag *et al.* 2016). The quality and quantity of DNA were measured on Multiskan spectrophotometer (Thermo Scientific). Further quality of bacterial DNA was checked in 0.7% agarose gel.

PCR-based detection of ARGs

ARGs were screened for each sample through simplex PCR amplification using ARG-specific primers. The primers used in this study were designed based on earlier studies (Table S4). PCR was performed in 15 μ l reaction volume using 100 ng template DNA, 10 pmol primer, and Q5 High-Fidelity DNA polymerase 2X Master Mix (NEB). PCR conditions were set as per the protocol provided with the enzyme, which includes, initial denaturation at 98°C for 30 s, followed by 35 cycles of denaturation (98°C for 10 s), annealing (30 s), extension (72°C) and a final extension at 72°C for 2 min. The annealing temperature and extension time were standardized for each set type of ARG primers. The presence or absence of gene was detected by electrophoresis on 1–2% agarose gel as per their expected amplicon size. In case of the presence of multiple bands in the agarose gel, the expected bands from randomly selected samples were excised, extracted from the gel, and subjected to Sanger sequencing for validation.

Statistical analysis

The Kruskal–Wallis test is applied to determine the significant differences between the three ethnic groups, Bhatra, Gond, and Paraja. We have compared individual genes across the three ethnic groups. Each violin plot represents the pairwise comparison between the ethnicity of the total frequency of an individual gene. The Wilcoxon rank test is applied to determine the significant differences between the alcoholic and nonalcoholic groups. This comparison has been done in two ways. First, the total number of gene has been taken across all the samples and compared between alcoholic and nonalcoholic categories. Second, we carried out an individual gene-wise comparison between alcoholic and nonalcoholic groups.

Acknowledgements

This work has been supported by the Department of Biotechnology, Government of India (BT/ILS/Flagship/2019). We sincerely acknowledge the technical support given by Sachikant Rout, Suraja Kumar Das and Santosini Dash of Institute of Life Sciences (ILS), Bhubaneswar. Manisha Sethi is a recipient of the University Grants Commission (UGC) fellowship, Government of India. Varshasata Mohanty is a recipient of the Women Scientist-A (WOS-A) award from the Department of Science and Technology (DST), Government of India. Graphical abstract has been created with BioRender.com.

Conflict of Interest

No conflict of interest has been declared.

Author contributions

Sethi M, Mohanty V, Parida A, Senapati S: The conception and design of the study. Sethi M, Mohanty V, Mishra S, Mahapatra S, Dash J, Parida D, Mohapatra AP, Mishra R, Prasad P, Parida A and Senapati S: Acquisition of data, or analysis and interpretation of data. Sethi M, Mohanty V, Mishra S, Mahapatra S, Dash J, Parida D, Mohapatra AP, Mishra R, Prasad P, Parida A and Senapati S: Drafting the article or revising it critically for important intellectual content. Sethi M, Mohanty V, Mishra S, Mahapatra S, Dash J, Parida D, Mohapatra AP, Mishra R, Prasad P, Parida A and Senapati S: Final approval of the version to be submitted.

References

- Angst, D.C., Tepekule, B., Sun, L., Bogos, B. and Bonhoeffer, S. (2021) Comparing treatment strategies to reduce antibiotic resistance in an in vitro epidemiological setting. *Proc Natl Acad Sci USA* **118**, 1–7.
- Bag, S., Ghosh, T.S., Banerjee, S., Mehta, O., Verma, J., Dayal, M., Desigamani, A., Kumar, P. *et al.* (2019) Molecular insights into antimicrobial resistance traits of commensal human gut microbiota. *Microb Ecol* **77**, 546–557.
- Bag, S., Saha, B., Mehta, O., Anbumani, D., Kumar, N., Dayal, M., Pant, A., Kumar, P. *et al.* (2016) An improved method for high quality metagenomics DNA extraction from human and environmental samples. *Sci Rep* **6**, 1–9.
- Bartoloni, A., Bartalesi, F., Mantella, A., Dell'Amico, E., Roselli, M., Strohmeyer, M., Barahona, H., Barrón, V. *et al.* (2004) High prevalence of acquired antimicrobial resistance unrelated to heavy antimicrobial consumption. *J Infect Dis* **189**, 1291–1294.
- Chng, K.R., Li, C., Bertrand, D., Ng, A.H.Q., Kwah, J.S., Low, H.M., Tong, C., Natrajan, M. *et al.* (2020) Cartography of opportunistic pathogens and antibiotic resistance genes in a tertiary hospital environment. *Nat Med* **26**, 941–951.
- Ciocan, D., Rebours, V., Voican, C.S., Wrzosek, L., Puchois, V., Cassard, A.M. and Perlemuter, G. (2018) Characterization of intestinal microbiota in alcoholic patients with and without alcoholic hepatitis or chronic alcoholic pancreatitis. *Sci Rep* **8**, 4822.
- Cockerill, F.R. (1999) Genetic methods for assessing antimicrobial resistance. *Antimicrob Agents Chemother* **43**, 199–212.
- Ding, T. and Schloss, P.D. (2014) Dynamics and associations of microbial community types across the human body. *Nature* **509**, 357–360.

- Ghosh, T.S., Sen Gupta, S., Bhattacharya, T., Yadav, D., Barik, A., Chowdhury, A., Das, B., Mande, S.S. *et al.* (2014) Gut microbiomes of Indian children of varying nutritional status. *PLoS One* **9**, e95547.
- Gueimonde, M., Sánchez, B., G. de los Reyes-Gavilán, C. and Margolles, A. (2013) Antibiotic resistance in probiotic bacteria. *Front Microbiol* **4**, 202.
- Haro, C., Rangel-Zúñiga, O.A., Alcalá-Díaz, J.F., Gómez-Delgado, F., Pérez-Martínez, P., Delgado-Lista, J., Quintana-Navarro, G.M., Landa, B.B. *et al.* (2016) Intestinal microbiota is influenced by gender and body mass index. *PLoS One* **11**, e0154090.
- Hasan, N. and Yang, H. (2019) Factors affecting the composition of the gut microbiota, and its modulation. *PeerJ* **7**, e7502.
- Hu, Y., Yang, X.I., Qin, J., Lu, N.A., Cheng, G., Wu, N.A., Pan, Y., Li, J. *et al.* (2013) Metagenome-wide analysis of antibiotic resistance genes in a large cohort of human gut microbiota. *Nat Commun* **4**, 1–7.
- Hu, Y., Yang, X., Lu, N. and Zhu, B. (2014) The abundance of antibiotic resistance genes in human guts has correlation to the consumption of antibiotics in animal. *Gut Microbes* **5**, 245–249.
- Karkman, A., Pärnänen, K. and Larsson, D.J. (2019) Fecal pollution can explain antibiotic resistance gene abundances in anthropogenically impacted environments. *Nat Commun* **10**, 1–8.
- Kim, Y.S., Unno, T., Kim, B.Y. and Park, M.S. (2020) Sex differences in gut microbiota. *World J Mens Health* **38**, 48–60.
- Milanović, V., Aquilanti, L., Tavoletti, S., Garofalo, C., Osimani, A., De Filippis, F., Ercolini, D., Ferrocino, I. *et al.* (2020) Distribution of antibiotic resistance genes in the saliva of healthy omnivores, ovo-lacto-vegetarians, and vegans. *Genes* **11**, 1088.
- Moriguchi, K., Zoolkefli, F.I.R.M., Abe, M., Kiyokawa, K., Yamamoto, S. and Suzuki, K. (2020) Targeting antibiotic resistance genes is a better approach to block acquisition of antibiotic resistance than blocking conjugal transfer by recipient cells: a genome-wide screening in *Escherichia coli*. *Front Microbiol* **10**, 2939.
- Ohashi, Y. and Fujisawa, T. (2017) Detection of antibiotic resistance genes in the feces of young adult Japanese. *Biosci Microbiota Food Health* 17–004.
- Rolain, J.-M. (2013) Food and human gut as reservoirs of transferable antibiotic resistance encoding genes. *Front Microbiol* **4**, 173.
- Van Schaik, W. (2015) The human gut resistome. *Philos Trans R Soc Lond B Biol Sci* **370**, 20140087.
- Sengupta, S., Chattopadhyay, M.K. and Grossart, H.-P. (2013) The multifaceted roles of antibiotics and antibiotic resistance in nature. *Front Microbiol* **4**, 47.
- Seville, L.A., Patterson, A.J., Scott, K.P., Mullany, P., Quail, M.A., Parkhill, J., Ready, D., Wilson, M. *et al.* (2009) Distribution of tetracycline and erythromycin resistance genes among human oral and fecal metagenomic DNA. *Microb Drug Resist* **15**, 159–166.
- Sinha, T., Vich Vila, A., Garmaeva, S., Jankipersadsing, S.A., Imhann, F., Collij, V., Bonder, M.J., Jiang, X. *et al.* (2019) Analysis of 1135 gut metagenomes identifies sex-specific resistome profiles. *Gut Microbes* **10**, 358–366.
- Sommer, M.O.A., Dantas, G. and Church, G.M. (2009) Functional characterization of the antibiotic resistance reservoir in the human microflora. *Science* **325**, 1128–1131.
- Souza, V., Rocha, M., Valera, A. and Eguiarte, L.E. (1999) Genetic structure of natural populations of *Escherichia coli* in wild hosts on different continents. *Appl Environ Microbiol* **65**, 3373–3385.
- Takagi, T., Naito, Y., Inoue, R., Kashiwagi, S., Uchiyama, K., Mizushima, K., Tsuchiya, S., Dohi, O. *et al.* (2019) Differences in gut microbiota associated with age, sex, and stool consistency in healthy Japanese subjects. *J Gastroenterol* **54**, 53–63.
- Van, T.T.H., Yidana, Z., Smooker, P.M. and Coloe, P.J. (2020) Antibiotic use in food animals worldwide, with a focus on Africa: Pluses and minuses. *J Global Antimicrob Resist* **20**, 170–177.

Supporting Information

Additional Supporting Information may be found in the online version of this article:

Table S1 Details of the individuals recruited for the study.

Table S2 Antibiotic resistance gene-type abundance in all samples based on the tribe.

Table S3 Antibiotic resistance gene-type abundance in all samples based on the alcohol consumption.

Table S4 List of primers and their sequences (forward and reverse) used in the current study.